

DESIGN DESCRIPTION:

1. Introduction

The design of the Smart Garbage Monitoring System revolves around creating an intelligent and automated solution for efficient waste management. The system is designed to continuously monitor the garbage level in bins and notify the concerned authorities when the bin is full. The design integrates electronic sensors, a microcontroller, and wireless communication modules to provide a real-time monitoring mechanism.

In this project, the main components used are ESP32 microcontroller, Ultrasonic Sensor, 9V Battery, and Jumper Wires. Each of these components plays a crucial role in achieving the desired functionality of the system. The design description below explains how each component is connected, configured, and utilized to ensure smooth and accurate operation.

2. ESP32 Microcontroller Design

The ESP32 serves as the core processing and control unit in the design of the Smart Garbage Monitoring System. It is responsible for interfacing with the ultrasonic sensor, processing the data, and transmitting the garbage level status through its Wi-Fi capability.

a. Hardware Design

The ESP32 board is mounted on a breadboard or a PCB base for easy connectivity. The board operates at 3.3V logic level, but it can be powered through the Vin pin (5V) from a regulated source such as a 9V battery through a voltage regulator.

The GPIO pins of the ESP32 are used to interface with the Ultrasonic Sensor:

- Trigger Pin (Digital Output): Sends a pulse to the ultrasonic sensor.
- Echo Pin (Digital Input): Receives the reflected signal from the sensor.

The board also includes an onboard Wi-Fi antenna, which is used to send real-time garbage status updates to a cloud platform or mobile application.

b. Software Design

The software part of ESP32 is programmed using the Arduino IDE or MicroPython, where a code is uploaded to continuously read distance data from the ultrasonic sensor. The ESP32 compares this data with predefined threshold levels (for example, 80% full) to determine the bin's status. When the garbage level crosses the threshold, the ESP32 sends an alert message or updates the status online.

This design ensures that the ESP32 efficiently manages sensor data processing, decision-making, and communication simultaneously, while consuming minimal power.

3. Ultrasonic Sensor Design

The Ultrasonic Sensor (commonly HC-SR04) is the main sensing unit used to detect the level of garbage inside the bin. Its design is based on non-contact distance measurement, making it ideal for this hygienic and robust application.

a. Working Principle

The ultrasonic sensor operates by transmitting ultrasonic waves at a frequency of 40 kHz through its transmitter (T). These waves travel through the air and are reflected back when they hit the surface of the garbage. The reflected waves are then received by the receiver (R).

The time taken for the signal to travel to the garbage and back is measured and used to calculate the distance using the formula:

$$\text{Distance} = \text{Time} \times \text{Speed of sound} / 2$$

This measurement helps determine the height of the empty space in the bin. By comparing this distance to the total height of the bin, the system can calculate the fill level percentage.

b. Design Integration

The sensor is placed at the top inner portion of the dustbin lid, facing downward toward the garbage. Its two terminals, Trigger and Echo, are connected to the ESP32's GPIO pins using jumper wires. The VCC and GND pins of the sensor are connected to the 5V supply and ground, respectively.

This design allows the sensor to take continuous measurements and send data to the ESP32 without physical contact, ensuring that the system remains durable and easy to maintain.

c. Functional Design Role

Whenever garbage is added, the distance between the sensor and garbage surface decreases. The ESP32 processes this information and determines whether the bin is full. The use of the ultrasonic sensor in this design ensures high accuracy, reliability, and cost-effectiveness.

4. Power Supply Design (9V Battery)

A reliable power supply is crucial for the stable functioning of any electronic system. In this project, a 9V battery is used to provide electrical energy to the ESP32 and the ultrasonic sensor.

a. Electrical Design

The 9V battery provides DC voltage, which is stepped down to 5V or 3.3V (depending on the circuit) using an onboard voltage regulator within the ESP32 board. This regulated voltage is then distributed to the connected components.

The positive terminal of the battery is connected to the Vin pin of the ESP32, while the negative terminal is connected to GND.

b. Design Considerations

Since the Smart Garbage Monitoring System is often placed outdoors or in public locations, using a battery allows the system to operate independently of external power sources.

To improve efficiency, rechargeable 9V batteries or solar charging modules can be integrated into the design, making the system more sustainable and environmentally friendly.

c. Role in System Design

The power supply design ensures uninterrupted operation of the circuit, even in remote areas where electrical outlets are unavailable. The compact and lightweight design of the 9V battery also makes it easy to enclose within the garbage bin's housing.

5. Jumper Wires Design

Jumper wires act as the connective elements in the circuit design. They link the ESP32, Ultrasonic Sensor, and the battery to form a complete working system. Two main types of jumper wires are used in the design: Male-to-Male and Female-to-Male.

a. Physical Design

- Male-to-Male Wires: Used for connecting components on the breadboard during prototyping and testing.
- Female-to-Male Wires: Used for connecting modules like the Ultrasonic Sensor (which has male header pins) to the ESP32 (which has female headers).

b. Design Role

The use of jumper wires allows for easy assembly and disassembly of the system, which is beneficial during testing and debugging. Since no soldering is required, modifications can be made quickly without damaging components.

Proper wire management ensures a neat design layout, minimizes short circuits, and improves the system's overall stability.

6. System-Level Design Integration

In the final system design:

- The Ultrasonic Sensor is fixed on the lid of the garbage bin.
- The ESP32 board is placed inside a protective box, connected to the sensor through jumper wires.
- The 9V Battery powers the entire system.
- The ESP32 reads distance data from the sensor and sends status updates via Wi-Fi to a cloud dashboard or mobile app.

This integrated design ensures real-time monitoring of waste levels, low power consumption, and minimal human intervention, aligning perfectly with the goals of a smart city initiative.